Goal-Oriented Deep Fourier Residual Methods

Carlos Vera-Ciro

Independent Researcher,

Abstract

The Deep Fourier Residual (DFR) method of Taylor, Pardo, and Muga [1] minimizes the H^{-1} dual norm of the weak PDE residual, a loss that is equivalent to the H^1 energy-norm error for well-posed problems. In many applications, however, the goal is not the global error but a quantity of interest (QoI): a point value, a subdomain average, or a boundary flux. We introduce a goal-oriented DFR method that applies the dual-weighted residual (DWR) framework to the DFR dual-norm loss. A primal network and an adjoint network are trained together; the primal network minimizes a QoI-weighted residual functional $|\langle R(u_h), z_h\rangle|$, while the adjoint network is trained to solve its own DFR problem, so that it approximates the true adjoint rather than merely driving the pairing to zero. Our main theoretical result is a goal-oriented counterpart of the DFR error–loss equivalence: the QoI-weighted loss, plus a computable remainder, controls the error $|J(u) - J(u_h)|$ in the quantity of interest. Unlike the DFR estimate, the bound is one-sided; we verify it empirically and find it holds on every run, at a median of ten times the true error. We also find that the QoI-weighted loss must not be read as an error estimator: because training minimizes it directly, it falls five orders of magnitude below the error that remains, and the bound is carried entirely by its remainder. Empirically, we vary the difficulty of the problem directly, at a fixed network and a fixed discretization, and find that goal-orientation helps whenever the baseline's optimization has stalled. When the problem is too hard for the network, plain DFR's error grows by a factor of 84 while goal-orientation's grows by 11, and goal-orientation is 6 to 15 times more accurate. When the problem is easy but plain DFR stops at an accuracy floor insensitive to both problem difficulty and network capacity, goal-orientation passes that floor by a factor of 15. Between the two, on easy problems at small networks, it is 2.4 times worse than the baseline: the method can hurt. The organizing variable is not the difficulty of the problem but whether the baseline has stopped making progress. The theory predicts

neither behavior, and we say where it falls short.

Keywords: Deep Fourier Residual, physics-informed neural networks, dual-weighted residual, goal-oriented error estimation, quantity of interest

1. Introduction

Neural networks have been used to solve differential equations since at least the work of Lagaris et al. [2], their ability to represent solutions resting on classical universal-approximation results [3]. Physics-Informed Neural Networks (PINNs) [4] have since emerged as a popular paradigm for solving partial differential equations (PDEs), embedding physical constraints directly into the neural network training loss.

Taylor, Pardo, and Muga [1] introduced the Deep Fourier Residual (DFR) method, which uses the H^{-1} dual norm of the weak residual as the loss function. This approach provides several advantages over strong-form PINNs:

- The loss is equivalent to the H^1 error for well-posed problems
- Reliable error estimation during training
- Applicability to problems with limited regularity

1.1. Motivation

The DFR loss controls the *global* energy-norm error. In many applications, however, the object of interest is not the global error but a functional of the solution – a quantity of interest (QoI) such as a point value $u(x_0)$, a subdomain average, or a boundary flux. Driving the global error to a fixed tolerance can be wasteful when only the QoI matters, and conversely a small global error does not guarantee an accurate QoI.

The dual-weighted residual (DWR) framework [5] addresses exactly this in the finite element setting by introducing an adjoint solution that localizes the error with respect to the QoI, including the control of pointwise quantities [6]. More recently, machine learning has been used to weight finite-element residuals toward a quantity of interest [7, 8]; those methods are data-driven and require a known target functional, rather than adjoint-based in the DWR sense. Neural networks have also been used inside the DWR framework itself: Roth et al. [9] solve the adjoint problem with a network while keeping a classical finite element primal, and Chakraborty et al. [10] represent both the primal and the adjoint by networks for multigoal estimation. Both estimate the error of a discretization; neither uses the goal to define the training loss, and neither works in the DFR dual-norm setting. We bring DWR to the DFR dual-norm loss: the resulting goal-oriented loss is the QoI-weighted residual functional, evaluated with the same discretization DFR already uses for the H^{-1} norm.

1.2. Contributions

The main contributions of this work are:

1. A goal-oriented DFR method that applies the dual-weighted residual framework to the DFR H^{-1} dual-norm loss, with simultaneous training of a primal and an adjoint network.
2. A goal-oriented counterpart of the DFR error-loss equivalence: a theorem showing that the QoI-weighted dual-norm loss, plus a computable remainder, controls the error $|J(u) - J(u_h)|$ in the quantity of interest. The bound is one-sided, and we delimit what it does and does not claim.
3. An empirical verification of that bound, which holds on every run at a median of ten times the true error. The verification also shows that the QoI-weighted loss alone is not an error estimator: training drives it five orders of magnitude below the error that remains.
4. An empirical study that varies the difficulty of the problem directly, isolating the mechanism rather than inferring it from a comparison across dimensions. Goal-orientation helps in two distinct regimes – when the network is resolution-limited, and when the baseline has stalled at an accuracy floor despite spare capacity – and hurts between them. What unites the two is that plain DFR has stopped making progress, not that the problem is hard.

1.3. Organization

The remainder of this paper is organized as follows. Section 2 reviews the necessary background on PINNs and the DFR method. Section 3 presents the goal-oriented DFR method. Section 4 provides the theoretical analysis. Section 5 describes our numerical experiments, and Section 6 discusses the results. Finally, Section 7 concludes the paper.

2. Background

2.1. Problem Setting

We consider the abstract problem of finding $u \in U$ such that

$$\mathcal{R}(u) = 0, \tag{1}$$

where $\mathcal{R} : U \to V^*$ is the PDE operator mapping from a trial space U to the dual of a test space V.

For example, Poisson's equation with homogeneous Dirichlet boundary conditions:

$$-\Delta u = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{2}$$

can be expressed in weak form as: find $u \in H_0^1(\Omega)$ such that

$$\langle \nabla u, \nabla v \rangle_{L^2} = \langle f, v \rangle_{L^2} \quad \forall v \in H_0^1(\Omega). \tag{3}$$

2.2. Physics-Informed Neural Networks

Physics-Informed Neural Networks (PINNs) [4] approximate the solution using a neural network u_θ and minimize a loss function based on the PDE residual:

$$\mathcal{L}_{\text{PINN}}(\theta) = \frac{1}{N_f} \sum_{i=1}^{N_f} |\mathcal{R}^s(u_\theta)(x_i)|^2 + \frac{1}{N_u} \sum_{j=1}^{N_u} |u_\theta(x_j) - u(x_j)|^2, \tag{4}$$

where $\mathcal{R}^s$ denotes the strong-form residual, $\{x_i\}$ are collocation points, and the second term fits the prescribed initial and boundary data at $\{x_j\}$. Weighting the two terms against each other is a common later refinement.

2.3. Deep Fourier Residual Method

Variational (weak-form) PINNs replace the strong residual by a weak one tested against a chosen set of functions [11], and a posteriori error analyses are available in some settings [12]. Any such method must integrate the weak forms numerically, and the resulting quadrature error is itself a source of inaccuracy that can dominate the discretization [13]. The DFR method [1] takes this further and minimizes the H^{-1} norm of the weak residual:

$$\mathcal{L}_{\text{DFR}}(\theta) = \|\mathcal{R}^w(u_\theta)\|_{V^*}^2 . \tag{5}$$

For a box domain with homogeneous Dirichlet conditions, the sine functions ϕ_k are the L^2-orthonormal eigenfunctions of the Dirichlet Laplacian, with eigenvalues $\lambda_k = \sum_i (\pi k_i / L_i)^2$. Taking the test space $V = H_0^1(\Omega)$ with the energy inner product $\langle u, v \rangle_V = \langle \nabla u, \nabla v \rangle_{L^2}$, the dual norm of the residual is computed from its sine coefficients $\hat{R}_k$ by

$$\|R\|_{V^*}^2 = \sum_k \frac{|\hat{R}_k|^2}{\lambda_k}, \tag{6}$$

which the Discrete Sine Transform evaluates efficiently. In $d \geq 2$ the eigenvalue couples the axes through the *sum* of squared frequencies; it is not the product of the one-dimensional weights.

We depart here from [1] in one respect. That work equips V with the full H^1 inner product, so that its spectral weights carry an additive constant, $\lambda_k = 1 + \sum_i (\pi k_i / L_i)^2$. We use the energy inner product instead. The two norms are equivalent on $H_0^1(\Omega)$ by the Poincare inequality, so the choice does not affect well-posedness; we prefer the energy inner product because it makes the model problem's stability constants exactly $\gamma = M = 1$ (Section 4), which removes them from the error bounds. We write $\|\cdot\|_{V^*}$ rather than $\|\cdot\|_{H^{-1}}$ to keep the distinction visible.

In practice the sum in (6) is truncated at a finite mode count K per axis, and the coefficients $\hat{R}_k$ are themselves computed by quadrature. Both approximations are inherited from DFR, and both matter for the error bounds of Section 4: since the truncated sum omits non-negative terms, it *underestimates* the dual norm, so a computed bound is a true bound only insofar as the discarded tail is negligible. We return to this point in Section 4.

2.3.1. Key Properties

The DFR method enjoys the following properties:

Proposition 2.1 (Error equivalence [1]). *For well-posed problems, there exist constants $\gamma, M > 0$ such that*

$$\frac{1}{M} \|\mathcal{R}(u)\|_{V^*} \leq \|u - u^*\|_U \leq \frac{1}{\gamma} \|\mathcal{R}(u)\|_{V^*} . \tag{7}$$

This error equivalence ensures that minimizing the DFR loss is equivalent to minimizing the error, providing reliable training dynamics.

2.4. From global error to quantities of interest

The error equivalence above controls the *global* energy-norm error $\|u - u^*\|_U$. When the object of interest is a functional $J(u)$ of the solution, this global control is indirect: a small DFR loss guarantees a small H^1 error, but not necessarily an accurate $J(u)$ at a given computational budget. The dual-weighted residual framework [5] closes this gap by introducing an adjoint solution that relates the QoI error to the residual, which is the starting point for the method developed in Section 3.

3. Methodology

In this section we present the goal-oriented DFR method.

3.1. Overview

Let u^* solve the weak problem $a(u^*, v) = \ell(v)$ for all $v \in V$, with residual $R(u)(v) = \ell(v) - a(u, v)$. The DFR method approximates u^* by a network u_θ and minimizes $\|R(u_\theta)\|^2_{V^*}$. We instead target a quantity of interest $J : V \to \mathbb{R}$. We introduce the adjoint problem and an adjoint network, and minimize a QoI-weighted residual functional.

3.2. Goal-oriented loss

For a (linear) QoI J, the adjoint solution $z^* \in V$ solves

$$a(v, z^*) = J(v) \quad \forall v \in V. \tag{8}$$

The error in the QoI admits the representation

$$J(u^*) - J(u_\theta) = a(u^* - u_\theta, z^*) = R(u_\theta)(z^*). \tag{9}$$

Approximating the adjoint by a network z_ϕ, we define the goal-oriented loss

$$\mathcal{L}_{\mathrm{QoI}}(\theta, \phi) = |R(u_\theta)(z_\phi)|, \tag{10}$$

and a regularized objective that also keeps the adjoint network close to the true adjoint,

$$\mathcal{L}(\theta, \phi) = \mathcal{L}_{\mathrm{QoI}}(\theta, \phi) + \alpha \, \|R^*(z_\phi)\|^2_{V^*} + \beta \, \|R(u_\theta)\|^2_{V^*} \, , \tag{11}$$

where $R^*(z)(w) = J(w) - a(w, z)$, $w \in U$, is the adjoint residual, and $\alpha, \beta \geq 0$ are weights. The dual pairing $R(u_\theta)(z_\phi)$ and the dual norms are evaluated with the DFR discretization (Section 2). Our implementation ties the two weights, $\alpha = \beta$, which is the setting reported throughout; the experiments of Section 5 use $\alpha = \beta = 1$ in one dimension and $\alpha = \beta = 5$ in two, and Section 6.1 reports the 1D control at both.

A subtlety in the joint training deserves emphasis. The goal term $|R(u_\theta)(z_\phi)|$ must update only the *primal* network. If instead the adjoint parameters ϕ were allowed to descend this term, the optimizer could drive the pairing to zero trivially by making z_ϕ orthogonal to the current residual, without z_ϕ approximating z^* at all – a degenerate solution that destroys the localization the method relies on. We therefore hold the adjoint fixed in the goal term (a stop-gradient), so that z_ϕ is trained *solely* by its own DFR residual $\|R^*(z_\phi)\|^2_{V^*}$ and genuinely approximates the true adjoint, while u_θ is guided by the QoI-weighted residual.

The stop-gradient closes the adjoint side of this degeneracy but not the primal side, and it is worth being precise about what the goal term then does.

Since $R(u_\theta)(z_\phi)$ is a single scalar, the primal network can zero it without approximating u^*: the goal term enforces one orthogonality condition, not accuracy. What makes it useful is that $|\cdot|$ has a non-vanishing gradient at the origin, so the term acts as a persistent constraint $R(u_\theta)(z_\phi) \approx 0$ rather than a penalty that switches off once satisfied. This is the surrogate, in a setting with no discrete test space, for the Galerkin orthogonality that a finite element method would enjoy against its whole test space, and by Proposition 4.2 it is exactly what reduces the QoI error to the second-order remainder. Accuracy itself must come from the β term. The two roles are complementary, and the weights are not interchangeable: β buys energy accuracy and the goal term redirects it.

3.3. Evaluation of the dual pairing

For a network u_θ satisfying the boundary conditions, the residual acts on a test function v as $R(u_\theta)(v) = \int_\Omega (f + \Delta u_\theta)\, v\, dx$. The goal-oriented pairing is therefore the L^2 integral

$$R(u_\theta)(z_\phi) = \int_\Omega \big(f + \Delta u_\theta\big)\, z_\phi\, dx, \tag{12}$$

which we evaluate directly on the same quadrature grid already used to assemble the DFR transforms, so no additional quadrature is introduced. Equivalently, by Parseval's identity in the orthonormal sine basis, this equals $\sum_k \hat{R}_k \hat{z}_k$; we use the physical-domain form to avoid transforming the adjoint network.

3.4. Algorithm

Algorithm 1 Goal-Oriented DFR

Require: Primal network u_θ, adjoint network z_ϕ, QoI J, weights α, β
Ensure: Trained parameters θ^*, ϕ^*

```
Initialize θ, φ; precompute the DST matrix and H^{-1} weights
for k = 1, ..., K do
    Evaluate residual coefficients R̂(u_θ) and R̂*(z_φ) by DST
    Primal loss L_u = |R(u_θ)(z̄_φ)| + β ||R(u_θ)||^2_{V*}   ▷ z̄_φ: adjoint held fixed
    Adjoint loss L_z = α ||R*(z_φ)||^2_{V*}
    θ ← θ − η ∇_θ L_u;    φ ← φ − η ∇_φ L_z
end for
return θ*, φ*
```

3.5. Boundary condition enforcement

For homogeneous Dirichlet conditions, both networks use the cutoff construction

$$u_\theta(x) = \varphi(x) \cdot \tilde{u}_\theta(x), \qquad z_\phi(x) = \varphi(x) \cdot \tilde{z}_\phi(x), \tag{13}$$

where φ vanishes on $\partial\Omega$ and $\tilde{u}_\theta, \tilde{z}_\phi$ are the raw network outputs.

4. Theoretical Analysis

We now prove that the goal-oriented loss controls the error in the quantity of interest. This is the goal-oriented counterpart of the DFR error–loss equivalence [1] and is the central theoretical result of the paper. We stress at the outset that it is one-sided: it is a reliability bound, not an equivalence. The DFR estimate (16) bounds the error both above and below, whereas no lower bound holds here. Taking z_h far from z^* makes $\mathcal{L}_{\mathrm{QoI}}$ positive while the QoI error may vanish, so the loss cannot be bounded by the error it controls. All proofs are elementary and given inline.

4.1. Setting and assumptions

Let U and V be Hilbert spaces (trial and test), $a : U \times V \to \mathbb{R}$ a bilinear form, and $\ell \in V^*$. The primal problem is to find $u^* \in U$ with

$$a(u^*, v) = \ell(v) \qquad \forall v \in V. \tag{14}$$

For $w \in U$ define the residual functional $R(w) \in V^*$ by $R(w)(v) = \ell(v) - a(w, v)$. Then $R(u^*) = 0$ and, subtracting (14),

$$R(w)(v) = a(u^* - w, v) \qquad \forall v \in V. \tag{15}$$

We assume:

(A1) *Boundedness:* $|a(w, v)| \le M \|w\|_U \|v\|_V$ for all $w \in U$, $v \in V$.

(A2) *Inf–sup stability:* there is $\gamma > 0$ such that

$$\inf_{0 \ne w \in U} \sup_{0 \ne v \in V} \frac{a(w, v)}{\|w\|_U \|v\|_V} \ge \gamma \qquad \text{and} \qquad \inf_{0 \ne v \in V} \sup_{0 \ne w \in U} \frac{a(w, v)}{\|w\|_U \|v\|_V} \ge \gamma.$$

Conditions (A1)–(A2) are the hypotheses of the theorem Ern and Guermond [14] name after Banach, Necas and Babuska (BNB), following Necas' 1962 generalization of the Lax–Milgram theorem and Babuska's introduction of the inf–sup condition into finite element analysis [15]; see [16] for the Banach-space residual-minimization setting used here. They make the primal problem (14) and its adjoint (below) well posed. For the symmetric coercive model problem (Poisson, $U = V = H_0^1(\Omega)$, $a(u, v) = \langle \nabla u, \nabla v \rangle_{L^2}$, energy norm $\|v\|_U^2 = a(v, v)$) they hold with $\gamma = M = 1$.

Under (A1)–(A2) the DFR error–loss equivalence holds [1]: by (15), (A1), and the first inequality of (A2),

$$\frac{1}{M}\,\|R(w)\|_{V^*} \;\le\; \|u^* - w\|_U \;\le\; \frac{1}{\gamma}\,\|R(w)\|_{V^*}\,, \qquad \|R(w)\|_{V^*} := \sup_{0 \neq v \in V} \frac{R(w)(v)}{\|v\|_V}. \tag{16}$$

4.2. Quantity of interest and adjoint

Let the quantity of interest be a bounded linear functional $J \in U^*$. The associated *adjoint problem* is to find $z^* \in V$ such that

$$a(w, z^*) = J(w) \qquad \forall w \in U, \tag{17}$$

which is well posed by (A1) and the second inequality of (A2). For $z \in V$ define the adjoint residual $R^*(z) \in U^*$ by $R^*(z)(w) = J(w) - a(w, z)$. Then $R^*(z^*) = 0$ and, subtracting (17),

$$R^*(z)(w) = a(w, z^* - z) \qquad \forall w \in U. \tag{18}$$

Remark 4.1. *For $d \geq 2$ the point-evaluation functional $u \mapsto u(x_0)$ is not bounded on H^1. We realize it by the mollification $J_\sigma(u) = \int_\Omega \eta_\sigma(x - x_0)\, u(x)\, dx$ with η_σ a normalized Gaussian of width σ; its Riesz representative η_σ lies in $L^2(\Omega) \subset H^{-1}(\Omega) = U^*$, so $J_\sigma \in U^*$, and $J_\sigma \to \delta_{x_0}$ as $\sigma \to 0$. In one dimension $H^1(\Omega) \hookrightarrow C^0(\bar{\Omega})$, so point evaluation is already bounded and mollification is not needed on theoretical grounds. We nevertheless mollify in one dimension as well, at the same σ used for the adjoint right-hand side, so that the 1D and 2D experiments measure the same functional and the 1D control is comparable to the 2D result.*

4.3. Error representation

Proposition 4.2 (Exact error representation)**.** *Let u^* solve* (14) *and z^* solve* (17)*. For every $u_h \in U$ and every $z_h \in V$,*

$$J(u^*) - J(u_h) = R(u_h)(z^*) = R(u_h)(z_h) + R(u_h)(z^* - z_h). \tag{19}$$

Proof. Taking $w = u^* - u_h$ in (17) and then using (15),

$$J(u^*) - J(u_h) = J(u^* - u_h) = a(u^* - u_h, z^*) = R(u_h)(z^*).$$

The second equality in (19) is the splitting $z^* = z_h + (z^* - z_h)$ together with the linearity of $R(u_h)$. □

The first term $R(u_h)(z_h)$ is computable from the two networks; it is the goal-oriented loss

$$\mathcal{L}_{\mathrm{QoI}}(u_h, z_h) := |R(u_h)(z_h)|. \tag{20}$$

The second term involves the unknown exact adjoint z^* and is bounded next.

4.4. Goal-oriented error control

Theorem 4.3 (QoI error control)**.** *Under (A1)–(A2), for every $u_h \in U$ and $z_h \in V$,*

$$\left|J(u^*) - J(u_h)\right| \;\le\; \mathcal{L}_{\mathrm{QoI}}(u_h, z_h) \;+\; \frac{1}{\gamma}\, \|R(u_h)\|_{V^*}\, \|R^*(z_h)\|_{U^*}\,. \tag{21}$$

Proof. By Proposition 4.2, (20), and the triangle inequality,

$$|J(u^*) - J(u_h)| \le \mathcal{L}_{\mathrm{QoI}}(u_h, z_h) + |R(u_h)(z^* - z_h)|.$$

For the remainder, the definition of the dual norm gives

$$|R(u_h)(z^* - z_h)| \le \|R(u_h)\|_{V^*}\, \|z^* - z_h\|_V\,.$$

It remains to control $\|z^* - z_h\|_V$ by the adjoint residual. Applying the second inf–sup inequality of (A2) to $z^* - z_h \in V$ and using (18),

$$\gamma\, \|z^* - z_h\|_V \;\le\; \sup_{0\ne w\in U} \frac{a(w,\, z^* - z_h)}{\|w\|_U} \;=\; \sup_{0\ne w\in U} \frac{R^*(z_h)(w)}{\|w\|_U} \;=\; \|R^*(z_h)\|_{U^*}\,.$$

Combining the three displays yields (21). □

The three quantities on the right-hand side of (21) are exactly the terms minimized in the training objective of Section 3: the goal-oriented loss $\mathcal{L}_{\mathrm{QoI}} = |R(u_h)(z_h)|$ and the primal and adjoint DFR dual norms $\|R(u_h)\|_{V^*}$ and $\|R^*(z_h)\|_{U^*}$, which DFR evaluates spectrally. The bound (21) is therefore a posteriori and computable from the two trained networks alone, and the goal-oriented objective is the quantity it controls.

Two caveats attach to the word computable. First, the dual norms are spectral sums truncated at K modes per axis and assembled by quadrature (Section 2); the truncated sum discards non-negative terms, so it underestimates $\|R(u_h)\|_{V^*}$, and (21) is a guaranteed bound only up to the discarded tail. It is a true bound under the standard saturation assumption that the tail is a fixed fraction of the total, which the mode counts used in Section 5 make reasonable but which we do not verify. Second, and more consequentially in practice, $\mathcal{L}_{\mathrm{QoI}}$ is itself minimized during training. We return to what that does to (21) in Remark 4.5 and measure it in Section 6.4.

Corollary 4.4 (Adjoint trained to tolerance)**.** *If the adjoint network satisfies* $\|R^*(z_h)\|_{U^*} \le \varepsilon$*, then*

$$|J(u^*)-J(u_h)| \;\le\; \mathcal{L}_{\mathrm{QoI}}(u_h, z_h)+\frac{\varepsilon}{\gamma}\;\|R(u_h)\|_{V^*} \;\le\; \mathcal{L}_{\mathrm{QoI}}(u_h, z_h)+\frac{M\varepsilon}{\gamma}\;\|u^* - u_h\|_U\,,$$

the last step by the left inequality of (16)*, which bounds* $\|R(u_h)\|_{V^*}$ *by* $M\,\|u^* - u_h\|_U$*.*

Remark 4.5 (Second-order structure, and what it does not say)**.** *The remainder in* (21) *is the* product *of the primal and adjoint dual-norm residuals. If both networks are trained to energy accuracy* $O(\delta)$*, that is* $\|R(u_h)\|_{V^*}, \|R^*(z_h)\|_{U^*} = O(\delta)$*, then the remainder is* $O(\delta^2)$*: driving the goal term to zero leaves a QoI error that is quadratic in the energy accuracy of the two networks. This product structure is the characteristic feature of dual-weighted residual methods.*

Four qualifications matter, and we state them because the natural reading of the previous paragraph overreaches on all four.

First, the $O(\delta^2)$ *statement presupposes that* δ *is small, that is, that both networks* are *well resolved in the energy norm. It therefore says nothing about a regime in which global resolution is out of reach: there* δ *is not small, and* $\gamma^{-1}\delta^2$ *is not small either. The bound is weakest exactly where a goal-oriented method is most wanted, and it does not by itself predict an advantage there. Section 6.3 finds the advantage there anyway, so the bound accommodates the result without explaining it.*

Second, the classical second-order gain rests on Galerkin orthogonality, which annihilates the first-order term $R(u_h)(z_h)$ *for every discrete test function. Here there is no discrete test space and no such orthogonality. Minimizing* $\mathcal{L}_{\mathrm{QoI}}$ *enforces a single scalar condition,* $R(u_h)(z_h) \approx 0$*, in place of orthogonality against a whole subspace. That is enough for the error representation to reduce to the second-order remainder, but it is a far weaker constraint, and it is satisfiable without* u_h *being accurate.*

Third, the entire analysis is approximation-theoretic and contains no model of the optimization. This is a real limitation rather than a formality: Section 6.3 finds that plain DFR's accuracy is frequently set by where its optimizer stops rather than by what its discretization can represent, and in that regime a bound in terms of best-approximation quantities describes something other than what is limiting the method. Nothing in (21) *distinguishes a residual that is large because the network cannot do better from one that is large because training stopped early.*

Fourth, and consequently, $\mathcal{L}_{\mathrm{QoI}}$ *must not be read as an error* estimator *for a network trained on it. The optimizer minimizes it directly; being one scalar condition, it is driven essentially to zero regardless of the error that* *remains. Section 6.4 measures this:* $\mathcal{L}_{\mathrm{QoI}}$ *settles around* 10^{-10} *while the QoI error sits near* 10^{-4}*, some six orders of magnitude apart. A quantity minimized to zero cannot report what error is left. The useful computable object is the whole right-hand side of* (21)*, which is carried by the remainder term and which we verify there.*

Remark 4.6 (Recovery of standard DFR)**.** *Taking the supremum of the error representation over all unit goals recovers global energy control: since* $\sup_{\|J\|_{U^*}=1} J(u^* - u_h) = \|u^* - u_h\|_U$*, the worst case over goals reproduces the DFR equivalence* (16)*. Standard DFR therefore controls every quantity of interest simultaneously, whereas goal-oriented DFR controls a single prescribed goal, which is why the latter can be cheaper for a fixed target accuracy.*

5. Numerical Experiments

We study the goal-oriented method against a plain DFR baseline on one- and two-dimensional Poisson problems, asking whether the QoI-weighted loss yields a more accurate quantity of interest at matched cost.

5.1. Setup

5.1.1. Problems

In one dimension we use two manufactured solutions with homogeneous Dirichlet conditions:

- **Smooth:** $u(x) = \sin(2x)$ on $(0, \pi)$, with $f = 4\sin(2x)$.
- **Peaked:** an arctan profile of steepness $k = 8$ on $(0, 1)$, $u(x) = \arctan(k(x - \frac{1}{2})) - \ell(x)$ with ℓ the affine correction enforcing $u(0) = u(1) = 0$; its gradient is concentrated near $x = \frac{1}{2}$.

In two dimensions we use a separable bump on the unit square $(0, 1)^2$, $u(x, y) = A(x)\,A(y)$ with A the boundary-corrected arctan profile of steepness $k = 8$ above; its gradient is concentrated along the lines $x = \frac{1}{2}$ and $y = \frac{1}{2}$, and $f = -\Delta u$ follows in closed form.

We use the word peaked rather than sharp advisedly. At $k = 8$ the profile's gradient varies by only a factor of about three between its peak and its median, and the same $k = 8$ profile appears in the 1D problem, where it is resolved without difficulty. The 2D solution is not intrinsically harder than the 1D one; what changes is the dimension, and with it the difficulty of the optimization at a given network size. We return to this in Section 6.5. The quantity of interest is a point evaluation $J(u) = u(x_0)$, mollified by a normalized Gaussian of width σ (Section 4): at $x_0 = \pi/4$ (1D smooth) and $x_0 = 0.65$ (1D sharp), and at $x_0 = (0.65, 0.65)$ in 2D. The width σ balances localization against spectral resolution: too small a σ makes J_σ approach a Dirac spike, whose high-frequency content the truncated sine basis cannot represent without a large mode count, while too large a σ blurs the target away from a point value. We use $\sigma = 0.04$ in 1D and $\sigma = 0.07$ in 2D, which resolve well at the mode counts employed here.

For reference, the 1D experiments use 60 sine modes and 150 quadrature points, 2000 Adam steps at learning rate 2×10^{-3}, and 400 LBFGS iterations; the 2D experiments use 16 modes and 32 quadrature points per axis, 3000 Adam steps at learning rate 3×10^{-3}, and 500 LBFGS iterations. The mode budgets are not the limiting factor in either case: the $k = 8$ profile's sine coefficients decay fast enough that 16 modes already capture all but a fraction 10^{-7} of its energy, and 60 modes all but 10^{-11}.

5.1.2. Varying the difficulty

Comparing one dimension against two confounds the difficulty of the problem with everything else that differs between the two settings, including the mode count, the quadrature and the step budget. Our principal experiment therefore varies the difficulty *within* a single setting: we sweep the steepness $k \in \{2, 4, 8, 16\}$ of the 2D bump at two fixed network sizes (105 and 337 primal parameters) and one fixed discretization, over ten seeds, so that k is the only variable.

The mode count for this sweep is fixed at 24 per axis with 48 quadrature points, chosen for the sharpest case rather than inherited from the sweep above. This matters: the relative energy the sine basis fails to represent grows from 8×10^{-8} at $k = 8$ to 10^{-3} at $k = 32$, so a sweep that held the mode count at 16 would make the test space, and not the network, the binding constraint at large k, and would measure mode truncation while appearing to measure resolution. At 24 modes the tail stays below 7×10^{-7} for every k in the sweep. We use two network sizes rather than one because the smaller is the more resolution-limited and the more favorable to goal-orientation, while the larger is where the sweep of Section 6.2 finds goal-orientation weakest; a behavior present at both cannot be an artifact of the size we chose.

5.1.3. Methods and training

The plain DFR baseline trains a single network against the H^{-1} residual loss. The goal-oriented method trains a primal and an adjoint network against the objective of Section 3. Both use the same primal-network architecture, the same Fourier test space, and the same schedule: Adam followed by an LBFGS polish, at the same step counts and learning rate. In two dimensions both methods use the identical full tensor-product DST discretization of the weak residual, so the comparison isolates exactly the effect of adding goal-orientation. The cutoff construction enforces the boundary conditions. We sweep the network width and repeat over ten random seeds in two dimensions and three in one.

Degrees of freedom throughout count the *primal* network. The goal-oriented method also trains an adjoint network of the same size, so at a given abscissa it carries about twice the parameters and twice the per-step cost of the baseline. We match on the primal network because the adjoint is auxiliary and is discarded once the primal is trained, but the comparison is not one of equal total cost, and we make no claim that it is.

As a correctness check, we verify that the discrete DFR loss vanishes at the exact solution of each of the three problems. The check is by construction independent of each problem's closed-form forcing: the Laplacian of the exact solution is taken by finite differences, so the test detects a mismatch between the manufactured solution and its forcing rather than reproducing it. On all three problems the loss at u^* falls at least eight orders of magnitude below its value at a one-percent perturbation of u^*, and it increases monotonically with the size of that perturbation.

5.1.4. Metrics

We report the error in the (mollified) quantity of interest, $|J_\sigma(u_h) - J_\sigma(u^*)|$, as a function of the primal-network degrees of freedom. Results and their interpretation are given in Section 6.

5.1.5. Reproducibility

Code is available in the accompanying repository, `https://github.com/caverac/hp-dfr`, with documentation at `https://caverac.github.io/hp-dfr/`; the figures are regenerated from saved sweep data by a single command.

6. Results and Discussion

We compare goal-oriented DFR against a plain DFR baseline throughout, at matched primal-network degrees of freedom. The metric is the error in the mollified point QoI, $|J_\sigma(u_h) - J_\sigma(u^*)|$, reported as a median over seeds with bands spanning the seed min–max.

Sections 6.1 and 6.2 sweep the network width in one and two dimensions. They establish that goal-orientation helps in the 2D setting and not the 1D one, but they cannot say why, because the two settings differ in more than the difficulty of the problem. Section 6.3 therefore varies the difficulty directly, within one setting, and it is the experiment the conclusions rest on. Section 6.4 tests the theorem against the errors actually committed. The picture that emerges is not the one we expected: goal-orientation helps when the baseline's optimization has stalled, which happens both when the problem is too hard and when it is easy enough that the baseline stops at its own accuracy floor, and it hurts in between.

6.1. One dimension: DFR saturates, and goal-orientation mostly pays for the attempt

Figure 1 reports the 1D results on the smooth and peaked problems. Plain DFR reaches its optimization floor – a QoI error of about 3×10^{-5} – already at the smallest networks, a few tens of parameters. Once the residual is near zero in the discrete dual norm, the DFR error–loss equivalence (16) makes the trial solution, and hence the quantity of interest, accurate everywhere. On the smooth problem the floor is flat to within a factor of two across the whole sweep, and if anything drifts upward with width; there is no resolution to be gained.

In this saturated regime goal-orientation has little error to remove, and at small networks it pays for the attempt. At $\alpha = \beta = 1$ it is 3.5 to 21 times *less* accurate than plain DFR at small and moderate sizes, where its two-network objective is harder to optimize, and it draws level only at the largest.

Because the 2D experiments below use a different loss weight, the 1D sweep is run at both. Raising the weight to $\alpha = \beta = 5$ improves the goal-oriented QoI error by up to six times, but it does not overturn the picture: goal-orientation wins 7 of 24 runs at $\alpha = \beta = 5$ against 4 of 24 at $\alpha = \beta = 1$, and loses the median at every size but the largest. The weight alone does not manufacture an advantage, which is what licenses comparing the two dimensions at all.

One detail here is easy to dismiss as noise and turns out not to be. At the largest network, $\alpha = \beta = 5$ does not merely reach parity but *overtakes* plain DFR on both problems, by 5 times on the smooth one and 2 on the peaked one. We flagged this initially as an oddity. Section 6.3 shows it is a reproducible effect with a mechanism behind it, and Section 6.5 returns to it. One dimension is therefore a weaker control than it first appears: it shows goal-orientation does not manufacture an advantage at small networks, but it does not show the absence of one everywhere.

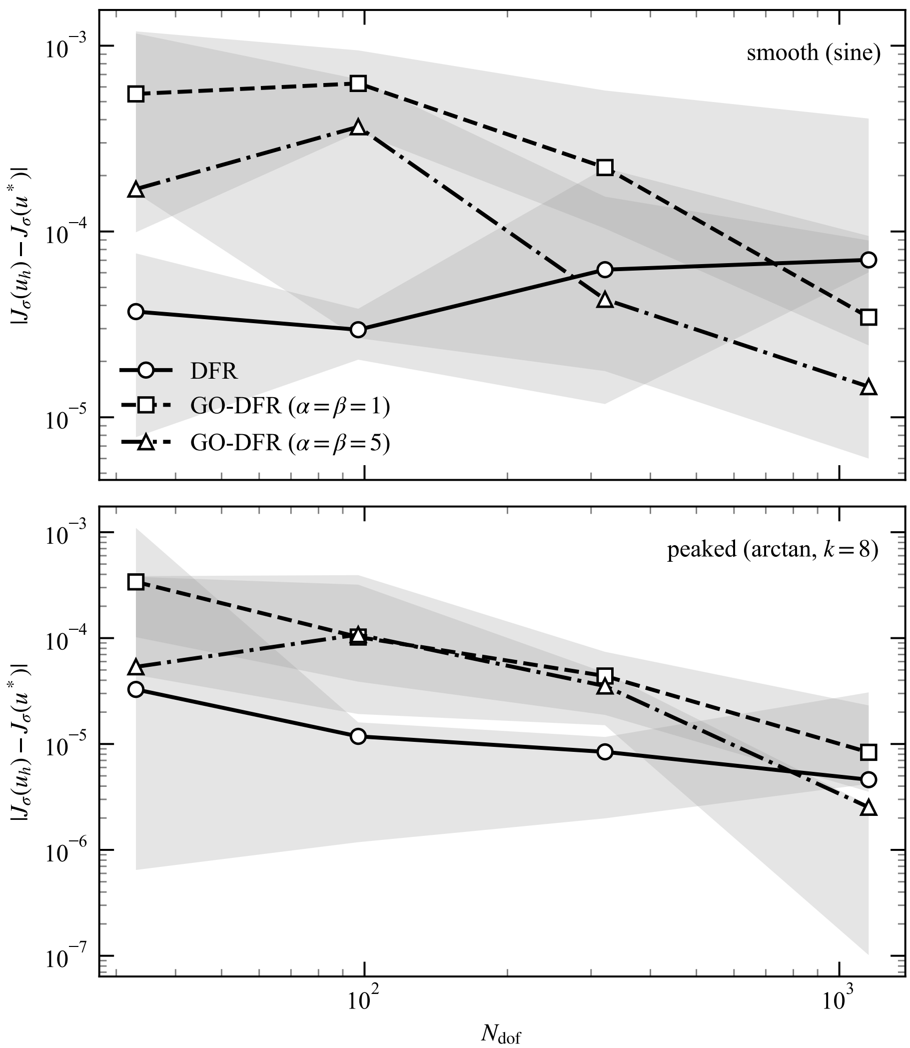

Figure 1: QoI error versus primal-network degrees of freedom (DOF) on 1D Poisson problems (smooth, top; peaked, bottom). Plain DFR reaches its optimization floor (about 3×10^{-5}) at extremely small network sizes and does not improve on it thereafter. Goal-oriented DFR (GO-DFR) is the worse method at small and moderate sizes, at either loss weight, so the weight used by the 2D experiments ($\alpha = \beta = 5$) does not by itself manufacture an advantage. At the largest networks, however, $\alpha = \beta = 5$ overtakes the baseline on both problems, passing a floor the baseline cannot: Section 6.3 identifies this as a reproducible effect rather than seed noise. Markers show medians over random seeds; bands span the seed min–max.

6.2. *Two dimensions: a goal-oriented advantage*

We repeat the comparison in two dimensions on the separable bump of Section 5. Both methods use the identical full tensor-product DST discretization, so the comparison isolates exactly the effect of adding goal-orientation. Unlike the 1D problems, a network of the same size does *not* resolve this solution everywhere: plain DFR sits at a median QoI error of 1.5×10^{-4} at the smallest networks and improves to around 10^{-5} as the width grows.

Figure 2 reports the outcome, and Table 1 gives the medians over ten seeds. Goal-orientation is the more accurate method at every size, winning 33 of the 40 runs. The median improvement is 2.3 to 3.9 times, with a geometric mean of 3.1, and it is remarkably flat across a twelvefold range of network sizes. Pooling the per-run paired ratios gives a median of 3.1 with a bootstrap 95% confidence interval of $[2.5, 6.8]$, so the effect is comfortably separated from unity in aggregate.

Ten seeds are necessary rather than decorative here, and it is worth saying why. The seed-to-seed spread of plain DFR at fixed size reaches a factor of 68; a three-seed median is a single middle draw from that, and the size-by-size ordering it produces is not stable. Our own earlier three-seed sweep put the 337 column at near-parity ($1.2\times$) and the 105 column at $5.3\times$, and both were sampling artifacts: at ten seeds they are $2.8\times$ and $3.9\times$. We report the per-size confidence intervals in the interest of not repeating the mistake: at 105, 697 and 1185 they exclude unity ($[2.8, 15.7]$, $[2.7, 18.0]$ and $[1.6, 2.9]$ respectively), whereas at 337 the interval is $[0.7, 19.2]$ and does not. No individual column of this table should be read as establishing a size-dependent trend; the aggregate should.

Neither method is monotone in the degrees of freedom, plain DFR being at its best at 697 and worse at 1185. Given the seed spread we do not attribute this to a mechanism either.

6.3. *Varying the difficulty directly*

The sweeps so far vary the network size at a fixed problem. They show *that* goal-orientation helps in two dimensions, but not that it helps *because* the network is resolution-limited: network size and problem difficulty are confounded in a single comparison against the 1D case. The steepness k separates them. Increasing k makes the same solution harder to represent at a fixed network and a fixed discretization, so it is the variable the mechanism is supposed to respond to, and it is the one we vary here. The mode count is held at 24 throughout, high enough that the sine basis captures even the sharpest case to a relative energy tail of 7×10^{-7}; holding it at the 16 of

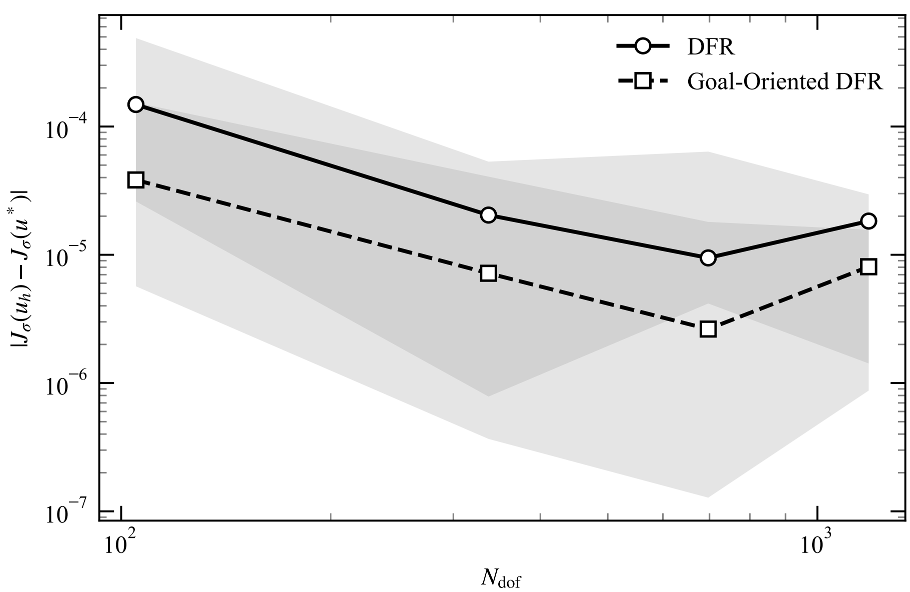

Figure 2: QoI error versus primal-network degrees of freedom (DOF) on the 2D Poisson problem with an arctan bump. In contrast to 1D, the 2D problem is resolution-limited: the network's global capacity is insufficient to drive the residual to zero everywhere. Under these conditions the goal-oriented DFR method (GO-DFR), which holds the adjoint fixed in the goal term and weights the dual norm by the coupled eigenvalues $\lambda_k = \sum_i (\pi k_i/L_i)^2$, redirects the network's capacity toward the support of the QoI. It wins 33 of the 40 runs and reduces the median QoI error by 2.3 to 3.9 times at matched primal-network DOF, a geometric mean of 3.1 times. Note that GO-DFR trains a second network of equal size, so it carries about twice the parameters at a given abscissa. Markers show medians over random seeds; bands span the seed min–max.

Table 1: Median point-QoI error on the 2D arctan bump, over ten random seeds, at matched primal-network degrees of freedom (DOF). Goal-oriented DFR (GO-DFR) improves the median QoI error at every size, by a geometric mean of 3.1 times, and wins 33 of the 40 runs. The improvement is flat across a twelvefold range of network sizes; given a seed-to-seed spread reaching a factor of 68, we do not read the size-to-size variation as a trend.

DOF	Plain DFR	GO-DFR	Improvement	GO-DFR wins
105	1.5×10^{-4}	3.8×10^{-5}	3.9×	8/10
337	2.0×10^{-5}	7.2×10^{-6}	2.8×	7/10
697	9.5×10^{-6}	2.6×10^{-6}	3.6×	9/10
1185	1.8×10^{-5}	8.1×10^{-6}	2.3×	9/10

the previous sweep would have made the test space rather than the network the binding constraint at large k, and the experiment would have measured mode truncation instead. Figure 3 reports the result at two network sizes, ten seeds each.

The prediction is confirmed, and the mechanism is visible in the curves rather than in any ratio. Plain DFR is flat while the problem is easy and then degrades sharply: at 105 degrees of freedom its median error grows by a factor of 84 between $k = 2$ and $k = 16$. Goal-oriented DFR degrades by a factor of 11 over the same range. The two curves cross near $k \approx 5$, and beyond it goal-orientation wins by 5.9 times at $k = 8$ and 11.1 at $k = 16$ (9/10 runs each, bootstrap 95% intervals $[2.2, 17.3]$ and $[6.3, 52.8]$). At 337 degrees of freedom the same hardening produces a 15.4-fold advantage at $k = 16$. This is the resolution-limited regime the method is for, isolated and measured rather than inferred.

The sweep also shows the method can *hurt*, which the network-size sweeps never made visible. At 105 degrees of freedom and $k = 4$ goal-orientation is 2.4 times *worse* than the baseline and wins only 2 of 10 runs: the problem is easy enough that there is no misallocated capacity to recover, and the second network makes the optimization harder for nothing. A practitioner choosing between the two methods needs this fact, and we would not have it from a sweep that varied only the network size.

The unexpected result is at the other end. At 337 degrees of freedom and $k = 2$ – the *easiest* problem in the study – goal-orientation is 15.4 times more accurate, winning 9 of 10 runs with a bootstrap interval of $[4.6, 40.0]$. This is not resolution limitation. Plain DFR's median error at this network size is 1.14×10^{-5}, 1.48×10^{-5} and 1.29×10^{-5} at $k = 2$, 4 and 8: flat to within a factor of 1.3 while the problem becomes four times harder. That is the signature of an optimization floor, not of insufficient capacity. Goal-orientation reaches 7.4×10^{-7} on the same problem, a factor of 15 *below* the floor the baseline cannot pass. It is also the same effect as the 1D overtake of Section 6.1, which we can now name.

So the sweep confirms the hypothesis it was designed to test and, at the same time, refutes the converse we had assumed. Goal-orientation is not useless when the global error is already resolved. Section 6.5 takes up what the two behaviors have in common.

6.4. Does the bound hold, and is the loss an estimator?

Theorem 4.3 is a posteriori and computable, so it can be checked rather than merely asserted. Every term on the right-hand side of (21) is evaluated at every training step; we record all three at the end of training for each of

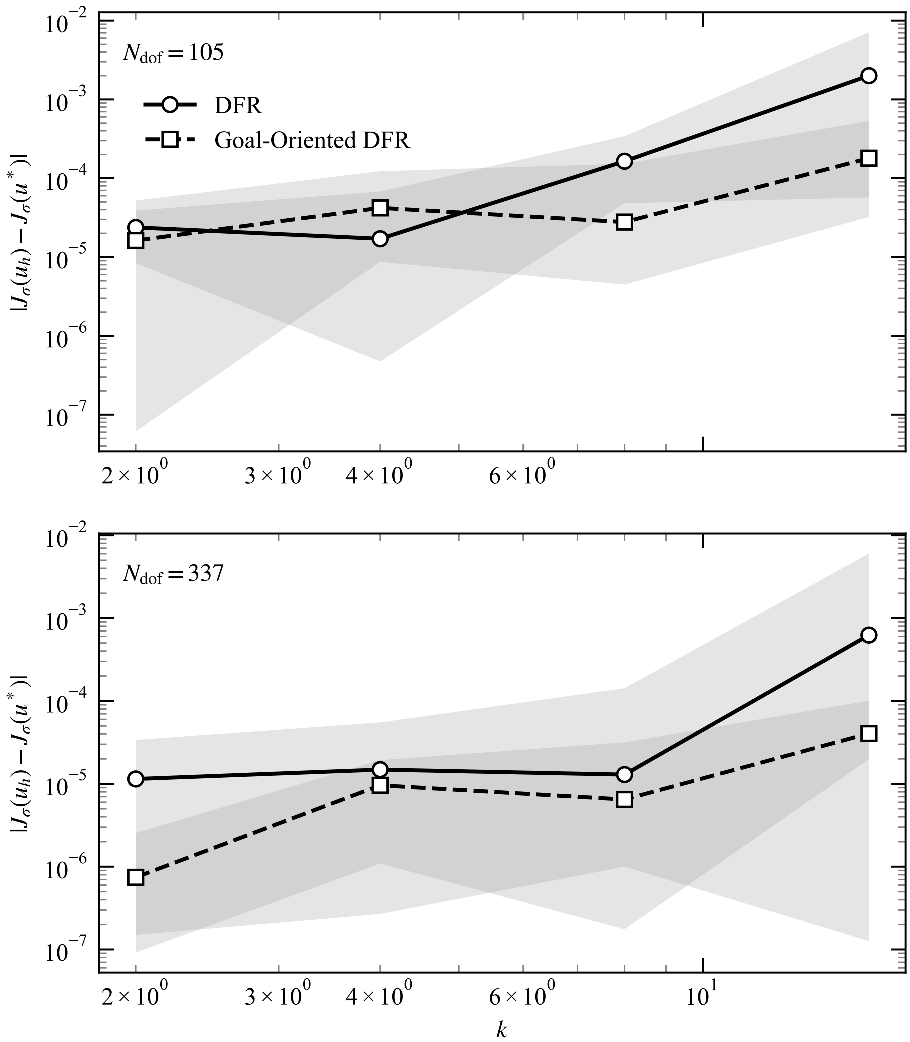

Figure 3: QoI error versus the solution's steepness k at fixed network size (105 degrees of freedom, top; 337, bottom) and fixed discretization, so that k is the only variable. Plain DFR sits at its optimization floor while the problem is easy and then degrades sharply, by a factor of 84 between $k = 2$ and $k = 16$ at 105 degrees of freedom; goal-oriented DFR degrades by a factor of 11 over the same range, so the gap opens as the problem hardens and the two curves cross near $k \approx 5$. Below the crossover, at $k = 4$ and 105 degrees of freedom, goal-orientation is the worse method. At 337 degrees of freedom it is ahead everywhere, including at $k = 2$, where the baseline is at its floor rather than resolution-limited. Markers show medians over ten random seeds; bands span the seed min–max.

the 168 goal-oriented runs of the sweeps above (48 in one dimension, 40 in two, and 80 in the steepness sweep), and compare their sum against the QoI error actually committed. Recall $\gamma = 1$ for this problem, so the bound is $\mathcal{L}_{\mathrm{QoI}} + \|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}$. Figure 4 shows the outcome.

Two things stand out, and they pull in opposite directions.

The bound holds, and is usefully tight. It is satisfied on all 168 runs, with no violation anywhere in the study, which spans an eightfold range of problem difficulty and a twelvefold range of network size. This is not a foregone conclusion: as noted in Section 4, the dual norms are evaluated on truncated spectral sums, which underestimate the true norms, so the computed right-hand side could in principle fall below the error. It does not, and by a comfortable margin: the bound exceeds the true QoI error by a median factor of 15, and never falls below 2.7. Within an order of magnitude, and never optimistic, is a useful a posteriori bound.

The goal-oriented loss is not an error estimator. The same data refutes the natural reading of $\mathcal{L}_{\mathrm{QoI}}$ as an estimate of the QoI error, and it is worth being blunt about it because the reading is inviting. Across all 168 runs $\mathcal{L}_{\mathrm{QoI}}$ has a median value of 2×10^{-10}, against a median QoI error near 10^{-5}: it is smaller than the quantity it is supposed to estimate by roughly five orders of magnitude, and it never once exceeds it. The reason is Remark 4.5: training minimizes $\mathcal{L}_{\mathrm{QoI}}$ directly, and being a single scalar condition it can be driven to zero without u_h being accurate. The optimizer does exactly that. Every part of the bound's value therefore comes from the remainder $\|R(u_h)\|_{V^*} \|R^*(z_h)\|_{U^*}$, whose median is 1.3×10^{-3} in one dimension.

This has a practical consequence worth stating plainly. A reader who took $\mathcal{L}_{\mathrm{QoI}}$ from a trained goal-oriented network and reported it as an error estimate would understate the error by five orders of magnitude. The estimator to use is the whole right-hand side of (21), and it is the remainder, not the goal term, that carries it. This is a general hazard of training on an a posteriori quantity, and not specific to DFR: any residual functional that appears in the loss is disqualified from also serving as an unbiased estimate of what the loss failed to remove.

6.5. Why the contrast, and what it means

Taken together the experiments show goal-orientation helping in two situations that look opposite and are not. In the first, the problem is too hard for the network: the residual cannot be made uniformly small, and the goal-oriented objective spends a limited capacity budget where the adjoint says it matters ($k \geq 8$ in Figure 3). In the second, the problem is easy and the network has capacity to spare, but plain DFR stops improving anyway, at

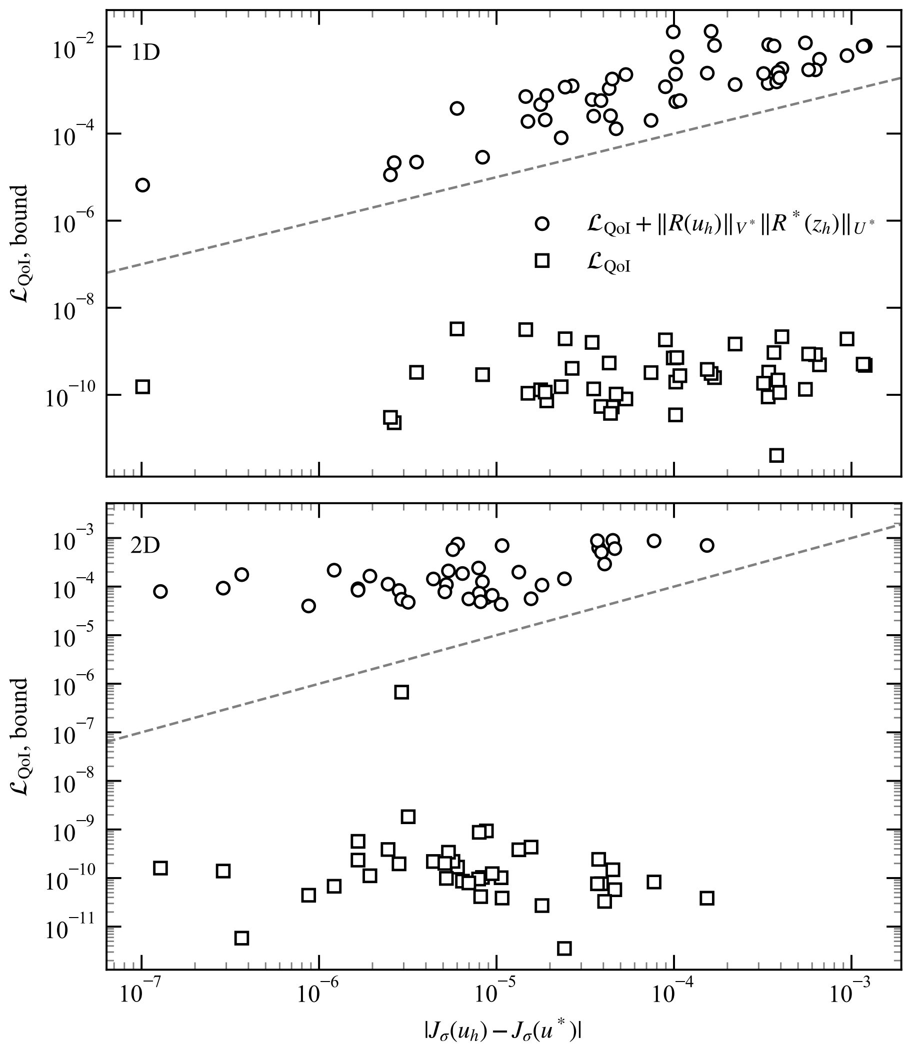

Figure 4: Verification of the QoI error bound (21) on the goal-oriented runs of the network-size sweeps (1D, top; 2D, bottom). The abscissa is the QoI error actually committed; the dashed diagonal is equality. The bound (circles) lies above the diagonal on every run, exceeding the true error by a median factor of 9.8 in 1D and 17.7 in 2D; across all 168 runs of the study, including the steepness sweep, it holds without exception. The goal term $\mathcal{L}_{\mathrm{QoI}}$ alone (squares) lies far *below* the diagonal: training minimizes it directly, driving it some five orders of magnitude beneath the error that remains, so it cannot serve as an error estimator and the bound is carried entirely by its remainder.

a floor near 10^{-5} that is insensitive to the difficulty of the problem ($k = 2$ at 337 degrees of freedom, and the largest 1D networks). Goal-orientation passes that floor by more than an order of magnitude.

What the two have in common is that plain DFR's optimization has stalled, for different reasons: against the capacity of the network in the first case, and against its own accuracy floor in the second. In both, the residual that remains is distributed without regard to the quantity of interest, and reweighting it toward the QoI recovers accuracy that plain DFR leaves on the table. Where goal-orientation loses – small networks on easy problems, most sharply $k = 4$ at 105 degrees of freedom – is where neither stall has set in: the baseline is still converging, there is nothing misallocated to recover, and the second network is a pure cost. The organizing variable is therefore not the difficulty of the problem but whether the baseline has stopped making progress, which is why the naive summary "goal-orientation helps when the problem is hard" is wrong at both ends of Figure 3.

We stress that Theorem 4.3 predicts neither behavior. It does not predict the resolution-limited gain, for the reason set out in Remark 4.5: the second-order remainder is small only when both networks are well resolved in energy, which is exactly what fails where that gain appears, so the bound is weakest where the method is most useful. It has still less to say about the second behavior, which is a statement about optimization – where an optimizer stops – and our analysis is entirely approximation-theoretic, with no model of the training dynamics at all. The mechanism we can point to is the one described in Section 3, that the goal term acts as a persistent orthogonality constraint redirecting the capacity budget toward the QoI, and no bound we prove quantifies it. Producing a theory that predicts rather than accommodates these results is the most interesting problem this work leaves open.

The floor itself deserves a comment, since it recurs throughout. Plain DFR's error is flat to a factor of 1.3 across a fourfold change in problem difficulty ($k = 2$ to 8 at 337 degrees of freedom), and flat to a factor of two across a twelvefold change in network size (Figure 1). A limit insensitive to both the difficulty of the problem and the capacity of the approximation is not an approximation limit; it is where the optimizer stops. We have not diagnosed its origin, and it is plainly worth doing: a method whose accuracy is set by its optimizer rather than its discretization is one whose error analysis, ours included, describes something other than what is limiting it in practice.

Two design points were essential to observing any of this. First, the adjoint network must be trained by its own residual and held fixed in the goal term (Section 3); otherwise it degenerates and the localization signal is lost.

Second, in two dimensions the dual-norm weights must use the true eigenvalues $\lambda_k = \sum_i (\pi k_i / L_i)^2$ rather than a separable product of one-dimensional weights, so that high-frequency residuals are penalized correctly.

6.6. Limitations

- The analysis assumes a linear QoI and a bounded, inf-sup-stable bilinear form.

- Theorem 4.3 is a reliability bound only. We prove no efficiency bound, and none holds without further assumptions on z_h, so the loss cannot be used to certify that the QoI error is large.

- The bound is stated for exact dual norms but evaluated on truncated spectral sums assembled by quadrature. It is therefore guaranteed only up to the discarded tail, which we argue is small at the mode counts used but do not bound rigorously.

- The seed-to-seed spread is large, reaching a factor of 68 at fixed network size, so individual columns of Table 1 carry real uncertainty even at ten seeds: the interval at 337 still includes unity. The aggregate is well separated from unity, but we make no claim about how the advantage varies with network size. The 1D control remains at three seeds, which is sufficient for a null result of its size but not for a fine reading of its columns.

- The 1D and 2D experiments differ in more than dimension: the mode count, the quadrature, the step budget and the learning rate all change between them. The steepness sweep of Section 6.3 avoids this confound entirely by varying difficulty within a single fixed setting, and it is the evidence we rely on for the mechanism; the cross-dimensional comparison should be read only as motivation.

- Goal-orientation is not uniformly beneficial: at 105 degrees of freedom and $k = 4$ it is 2.4 times worse than plain DFR, winning 2 of 10 runs. We report where it helps and where it does not, but we have no a priori test that tells a practitioner which regime a new problem is in.

- The accuracy floor that governs two of our three regimes is undiagnosed. We establish that it is insensitive to problem difficulty and network capacity, and therefore that it is a property of the optimization rather than the approximation, but we do not identify its cause. Our error analysis says nothing about it.

- All experiments use a single family of manufactured solutions (an arctan profile and its tensor product) with a mollified point QoI on a box. Other QoIs, other geometries and real applications remain untested.

- The goal term $|R(u_h)(z_h)|$ is non-differentiable at zero. Quasi-Newton optimizers such as L-BFGS build a Hessian approximation from successive gradients and are sensitive to this kink, so the LBFGS polish is less effective on the goal-oriented objective than on the smooth plain-DFR loss. Two smooth surrogates with the same minimizer are natural refinements: the squared pairing $\mathcal{L}^{\mathrm{sq}}_{\mathrm{QoI}} = R(u_h)(z_h)^2$, and a Huber-type smoothing $\mathcal{L}^{\varepsilon}_{\mathrm{QoI}} = \sqrt{R(u_h)(z_h)^2 + \varepsilon}$ that recovers $|R(u_h)(z_h)|$ as $\varepsilon \to 0$. Either would be expected to improve the quasi-Newton convergence rate; we did not require them to obtain the results above.

- The spectral evaluation of the dual norm requires a box (or box-decomposable) domain, as in the original DFR method; extending goal-orientation to re-entrant corner singularities would require pairing it with domain decomposition, as in the adaptive DFR method [17], and is left to future work.

7. Conclusion

We have presented a goal-oriented extension of the Deep Fourier Residual method. By introducing an adjoint network and minimizing a QoI-weighted dual-norm residual, the loss targets the error in a quantity of interest rather than the global energy norm, while reusing the DFR spectral machinery.

7.1. Summary

The main contributions are:

1. A goal-oriented DFR method coupling a primal and an adjoint network through a QoI-weighted dual-norm loss.
2. A goal-oriented counterpart of the DFR error-loss equivalence (Theorem 4.3): the QoI-weighted loss, plus a computable remainder, controls the error in the quantity of interest. The bound is one-sided, and it is weakest in the resolution-limited regime where goal-orientation is most useful.
3. An empirical verification of that bound, which holds on every run at a median of ten times the true error, and which shows that the QoI-weighted loss alone cannot serve as an error estimator for a network trained on it.

4. An empirical characterization of when goal-orientation helps, obtained by varying problem difficulty directly rather than inferring it across dimensions: it helps when the network is resolution-limited (6 to 15 times), helps when the baseline has stalled at its accuracy floor with capacity to spare (15 times), and hurts when neither holds (2.4 times worse). The theory predicts neither gain.

7.2. Future work

- **A bound that predicts the gain**: the most interesting gap this work leaves is that Theorem 4.3 is consistent with both observed advantages and explains neither, being weakest exactly where they appear. A bound that quantifies capacity reallocation, rather than presupposing global resolution, would be the natural next result.

- **Diagnosing the accuracy floor**: plain DFR stops at an error near 10^{-5} that is insensitive to both problem difficulty and network capacity, and which goal-orientation passes by an order of magnitude. Identifying what sets it would explain two of our three regimes and would bear on DFR generally, not only on its goal-oriented variant.

- **An analysis that sees the optimizer**: since the floor is a property of training rather than of approximation, error analysis in the best-approximation style cannot reach it. Bounds that account for where the optimizer actually stops are what the empirical picture calls for.

- **Nonlinear QoIs and PDEs**: extend the error representation to nonlinear functionals and operators.

- **Adaptive refinement**: combine the goal-oriented indicator with local refinement of the test space or trial networks.

- **General domains**: extend beyond box domains via domain decomposition.

7.3. Reproducibility

All code used in this paper is available in the accompanying repository, `https://github.com/caverac/hp-dfr`. The documentation site, `https://caverac.github.io/hp-dfr/`, gives instructions for reproducing the experiments and the figures.

Acknowledgments

The author thanks the MATHMODE group for making the original DFR implementations publicly available.

References

[1] J. M. Taylor, D. Pardo, I. Muga, A deep Fourier residual method for solving PDEs using neural networks, Computer Methods in Applied Mechanics and Engineering 405 (2023) 115850. doi:10.1016/j.cma.2022.115850.

[2] I. E. Lagaris, A. Likas, D. I. Fotiadis, Artificial neural networks for solving ordinary and partial differential equations, IEEE Transactions on Neural Networks 9 (5) (1998) 987–1000. doi:10.1109/72.712178.

[3] G. Cybenko, Approximation by superpositions of a sigmoidal function, Mathematics of Control, Signals and Systems 2 (4) (1989) 303–314. doi:10.1007/BF02551274.

[4] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.

[5] R. Becker, R. Rannacher, An optimal control approach to a posteriori error estimation in finite element methods, Acta Numerica 10 (2001) 1–102. doi:10.1017/S0962492901000010.

[6] S. Prudhomme, J. T. Oden, On goal-oriented error estimation for elliptic problems: application to the control of pointwise errors, Computer Methods in Applied Mechanics and Engineering 176 (1–4) (1999) 313–331. doi:10.1016/S0045-7825(98)00343-0.

[7] I. Brevis, I. Muga, K. G. van der Zee, A machine-learning minimal-residual (ML-MRes) framework for goal-oriented finite element discretizations, Computers & Mathematics with Applications 95 (2021) 186–199. doi:10.1016/j.camwa.2020.08.012.

[8] I. Brevis, I. Muga, K. G. van der Zee, Neural control of discrete weak formulations: Galerkin, least squares & minimal-residual methods with quasi-optimal weights, Computer Methods in Applied Mechanics and Engineering 402 (2022) 115716. doi:10.1016/j.cma.2022.115716.

[9] J. Roth, M. Schröder, T. Wick, Neural network guided adjoint computations in dual weighted residual error estimation, SN Applied Sciences 4 (2) (2022) 62. doi:10.1007/s42452-022-04938-9.

[10] A. Chakraborty, T. Wick, T. Rabczuk, X. Zhuang, Multigoal-oriented dual-weighted-residual error estimation using PINNs, Machine Learning for Computational Science and Engineering 1 (1) (2025) 13. doi:10.1007/s44379-025-00012-4.

[11] E. Kharazmi, Z. Zhang, G. E. M. Karniadakis, hp-VPINNs: Variational physics-informed neural networks with domain decomposition, Computer Methods in Applied Mechanics and Engineering 374 (2021) 113547. doi:10.1016/j.cma.2020.113547.

[12] S. Berrone, C. Canuto, M. Pintore, Solving PDEs by variational physics-informed neural networks: an a posteriori error analysis, Annali dell'Università di Ferrara 68 (2) (2022) 575–595. doi:10.1007/s11565-022-00441-6.

[13] J. A. Rivera, J. M. Taylor, Á. J. Omella, D. Pardo, On quadrature rules for solving partial differential equations using neural networks, Computer Methods in Applied Mechanics and Engineering 393 (2022) 114710. doi:10.1016/j.cma.2022.114710.

[14] A. Ern, J.-L. Guermond, Theory and Practice of Finite Elements, Vol. 159 of Applied Mathematical Sciences, Springer, New York, NY, 2004. doi:10.1007/978-1-4757-4355-5.

[15] I. Babuška, Error-bounds for finite element method, Numerische Mathematik 16 (4) (1971) 322–333. doi:10.1007/BF02165003.

[16] I. Muga, K. G. van der Zee, Discretization of linear problems in Banach spaces: residual minimization, nonlinear Petrov–Galerkin, and monotone mixed methods, SIAM Journal on Numerical Analysis 58 (6) (2020) 3406–3426. doi:10.1137/20M1324338.

[17] J. M. Taylor, M. Bastidas, V. M. Calo, D. Pardo, Adaptive deep Fourier residual method via overlapping domain decomposition, Computer Methods in Applied Mechanics and Engineering 427 (2024) 116997. doi:10.1016/j.cma.2024.116997.